

Closed-Domain Sanskrit Mahabharata System

Minimal-Footprint Translation Engine & Strictly Guardrailed Q&A Architecture

Executive Summary: This specification details the end-to-end architecture for processing the unabridged Sanskrit Mahabharata using minimal-footprint neural models. The system features an offline lightweight translation pipeline paired with a strictly quarantined Retrieval-Augmented Generation (RAG) Q&A engine. The chatbot is mathematically constrained to answer queries exclusively based on the Mahabharata corpus, actively refusing out-of-domain entities (such as Ramayana characters, modern pop-culture figures like Jon Snow or Harry Potter, and general world knowledge).

1. READILY AVAILABLE DATASETS

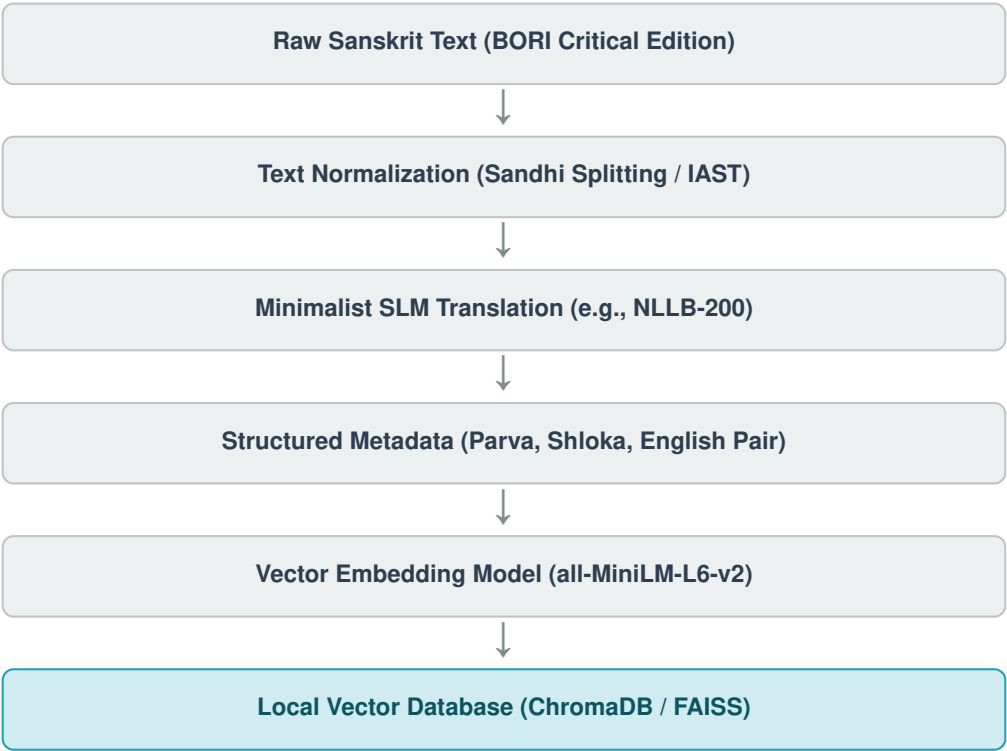
DATASET NAME	SOURCE / REPOSITORY	FORMAT & CONTENT	ARCHITECTURAL ROLE
BORI Critical Edition	GRETIL & SanskritDocuments	18 Parvas, ~80,000 verses in IAST	Primary canonical Sanskrit source text.
K. M. Ganguli Translation	Project Gutenberg	Full English prose translation	Parallel corpus alignment for English semantic grounding.
IIT Bombay Corpus	IITB NLP Repository	Parallel aligned sentence pairs	Base linguistic bootstrapping for the translator model.

2. END-TO-END SYSTEM PIPELINE & FLOW DIAGRAMS

To eliminate computational overhead and runtime latency, the architecture is decoupled into two independent stages: an **Offline Translation Layer** and an **Online Guardrailed Q&A Layer**.

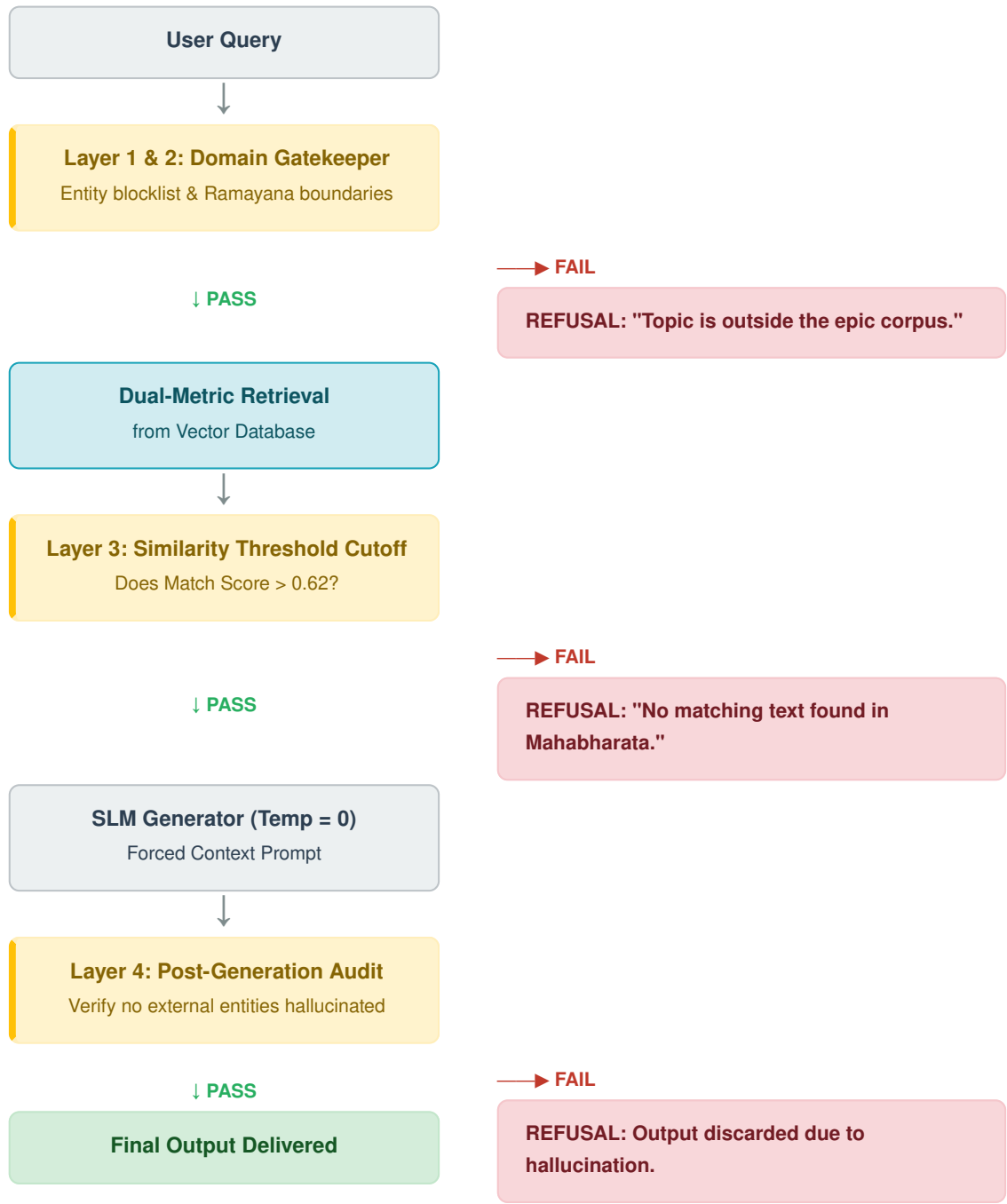
STAGE 1: OFFLINE INGESTION, TRANSLATION & INDEXING PIPELINE

This is a one-time execution pipeline to construct the foundational knowledge base.



STAGE 2: ONLINE GUARDRAILED QUERY-RESPONSE PIPELINE

This diagram maps out the multi-layered defense system, intercepting out-of-domain queries.



3. ULTRA-SMALL MODEL SELECTION

SYSTEM ROLE	MODEL CANDIDATE	SIZE / FOOTPRINT	OPERATIONAL PROFILE
Sanskrit Translator	NLLB-200 Distilled	600 Million	Runs offline once; zero runtime footprint.
Vector Embedding	all-MiniLM-L6-v2	22M (~90 MB)	Sub-millisecond retrieval speeds.
Domain Gatekeeper	Deterministic Trie Classifier	< 5M Parameters	Pre-retrieval boundary enforcement.
Response Generator	SmoLM2-135M (4-bit)	135M (~120 MB)	Executes strict synthesis on edge devices.

4. GUARDRAIL BEHAVIOR & TEST SCENARIOS

USER QUERY	ACTIVE DEFENSE LAYER	SYSTEM DECISION	OBSERVED BEHAVIOR
"What was Karna's vow?"	Layers 1, 2, 3 Cleared	PERMITTED	Synthesizes factual answer with citations.
"Who killed Jon Snow?"	Layer 1 (Entity Gatekeeper)	REFUSED	Blocked instantly: <i>Topic is outside corpus.</i>
"What spells did Harry Potter use?"	Layer 1 & Layer 3	REFUSED	Fails vector similarity test.
"Explain Ram vs Ravana."	Layer 2 (Disambiguation)	BOUNDED	Restricts response solely to Markandeya's narration in Vana Parva.